

BIOCLEAN · SCHOOL & COMMUNITY INNOVATION

BIOCLEAN SMART BIOGAS SYSTEM (BSBS)

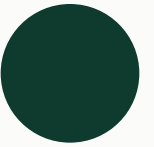
Combating Air Pollution in Schools and Communities in Enugu State
Creating a healthy environment for better learning and healthier living.

Presented by

Chinwuba Chinwendu Jennifer

Head Girl · **WASTE → ENERGY** College, Enugu

Introduction to Air Pollution



Air pollution is the release of substances into the atmosphere in quantities that can harm human health, modify

01 PM2.5 & PM10

Tiny dust particles from vehicles and generators; the most dangerous because they reach our lungs.

02 Carbon Monoxide (CO)

From the incomplete burning of fuel such as coal, firewood, diesel and petrol.

03 Sulphur Dioxide (SO₂)

From gas flaring and diesel.

04 Nitrogen Dioxide (NO₂)

A reddish-brown gas from vehicle engines, generators and gas-flaring industries.

05 Ground-level Ozone (O₃)

A major part of smog, formed when sunlight reacts with other pollutants.

06 Methane (CH₄)

A colourless, odourless gas from sewage, decomposing waste and gas flaring; forms ground-level ozone in sunlight.

WHO reports about **6.7 million premature deaths every year** caused by air pollution.

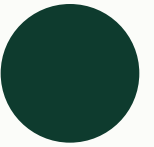
Most days in my school — especially within the hostels — my fellow students complain of drowsiness and nausea. This is due to the presence of ammonia (NH_3) and hydrogen sulphide (H_2S) gas from sewage and open dumps.

Students report:

- Drowsiness and nausea, especially within the hostels.
- Discomfort when inhaling smoke that comes from burning waste.
- Some students avoid entering classrooms because they cannot breathe comfortably.

Clean air is not a luxury — it is a learning condition.

My Community Challenge



Nowadays, most homes in my community utilise charcoal, stoves and generators for cooking and powering their houses. As we all know, coal burns in a limited supply of air to release carbon monoxide (CO), a colourless and odourless gas which, when inhaled excessively, accumulates in the body and reacts with the haemoglobin in our blood to form carboxyhaemoglobin. This cuts down the level of oxygen delivered in the body, leading to severe cardiovascular issues or even death.

The community need

Safer waste handling + cleaner energy + better sanitation → healthier, more productive homes and schools.



01 Health effects

Air pollution affects our health through diseases like asthma, pneumonia and lung cancer.

in

02 Academic effects

Students find it difficult to learn in polluted environments, leading to poor grades; teachers may also lose interest in teaching.

03 Economic effects

Families spend income on treatment and medical expenses, while illness reduces workers' productive days.

04 Social effects

Smoke, smog and bad odour reduce interaction; students face embarrassment and stigma over fears of airborne disease.

The Research and Science Behind the Innovation: Bioclean Smart Biogas System (BSBS)

What is biogas?

Biogas is a renewable fuel produced when microorganisms break down waste without oxygen.

The process



Comparison table (type your content here)

The Bioclean Smart Biogas System — How It Works

**INSERT BIOGAS SYSTEM
DIAGRAM / PICTURE HERE**

Drop in the biogas system diagram or photo.

A visual overview of how organic waste becomes clean, renewable energy.

Project Cost and Funding Plan

LILAN

¥1,000,000

School Management, P.T.A. &
CSR Partners

¥1,650,000

Total

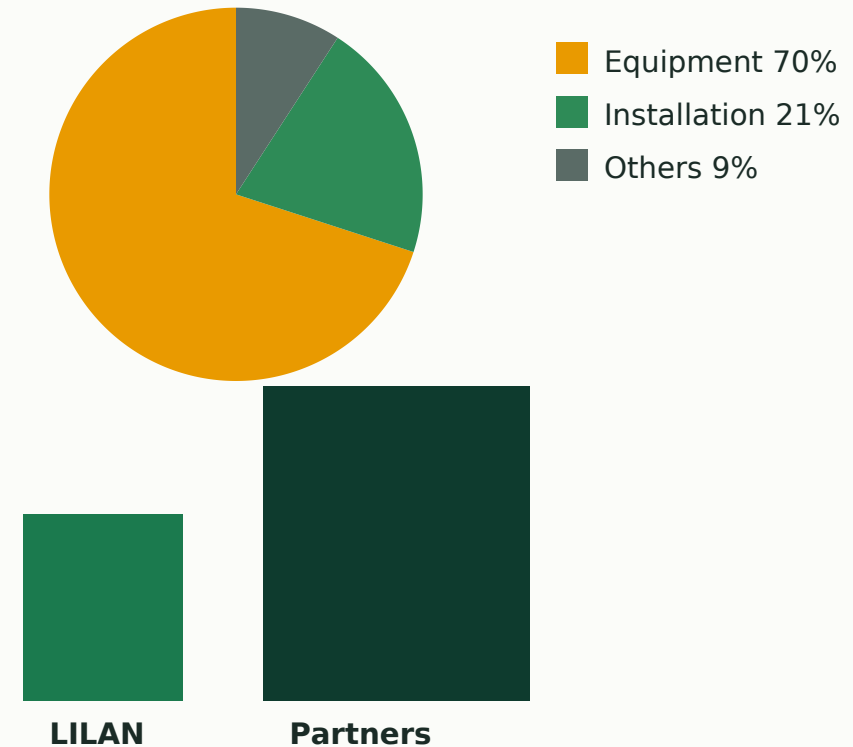
¥2,650,000

Motto

This investment pays for itself in 17 months and generates savings for 15 years.

Cost allocation

LILAN ¥1.0m vs Partners ¥1.65m
LILAN ¥1.0m vs Partners ¥1.65m



Financial Viability and Sustainability

Annual net benefit to school

#1,880,000

#1,860,000 Savings

No LPG and no waste-truck fees.

#320,000 Revenue

Organic fertilisers and carbon credits.

#80,000 Running cost

Maintenance and stipends.

#1,880,000 Net benefit to school

71%

Months to pay back

Rs. 1,880,000

Key Metrics & Sustainability

Payback period

17 months

ROI Year 1

71%

5-year savings

₹6,750,000 for our labs, infrastructure and scholarships.

As the Head Girl of my college, I will sustain this innovation by:

- 01** Organise 10 reliable students from our Green Club to manage its operations.
- 02** Train two (2) juniors for continuity.
- 03** Enhance scalability so more models can be replicated in other schools and communities.

THE SOLUTION

How My Smart Bioclean Biogas System (BSBS) Combats

BEFORE

Health

Biogas burns clean and releases no smoke, reducing risks from firewood, charcoal and generator fumes.

Agricultural

The slurry residue is collected and dried, then sold to farmers — fostering agriculture.

Environmental

BSBS traps methane (a greenhouse gas) and offers an alternative to firewood and charcoal.

AFTER

Economic

Biogas comes from decomposing organic materials, so BSBS is affordable for most families and schools.

Social

Students are no longer isolated or stigmatised; clean air for everyone, with BSBS in every home.

Transparency

App-based monitoring ensures accountability.

THANK YOU

Conclusion and Gratitude

01 The problem is real

300 students suffer daily from gas fumes; waste causes bad odour and ₦1.8m is wasted every year.

02 The solution works

Biogas turns our biggest problem into our biggest asset: clean cooking, clean hostels, plus ₦1.8m net gain every year.

03 The plan is ready

₦2.65m total cost: ₦1m from LILAN, ₦1.65m from partners; 17-month payback and 15 years of impact.

04 The leadership is here

With the BSBS Green Club and annual training, this impact will outlive me and inspire other schools in Enugu.

Invest ₦1,000,000 in me today. In return, LILAN gets a national model of student-led climate action that saves money, saves the planet and saves lives.

This is not just a project — this is my legacy.

Let's Breathe In Clean Air together.

Thank you, LILAN, for this opportunity to change my school, community, nation, continent and the whole world.
BIOCLEAR SMART BIOGAS SYSTEM · BSBS